

Figure 5: Parallel solution sets of $Ax=b$ and $Ax=0$

Theorem 6: Suppose the equation $Ax=b$ is consistent for some given b , and let p be a solution. Then the solution set of $Ax=b$ is the set of all vectors of the form $w=p+v_h$, where v_h is any solution of the homogeneous equation $Ax=0$.

Theorem 6 says that if $Ax=b$ has a solution, then the solution set is obtained by translating the solution set ~~is obtained by translating the~~ of $Ax=0$, using any particular solution p of $Ax=b$ for the translation.

• Figure 6 illustrates the case in which there are two free variables.

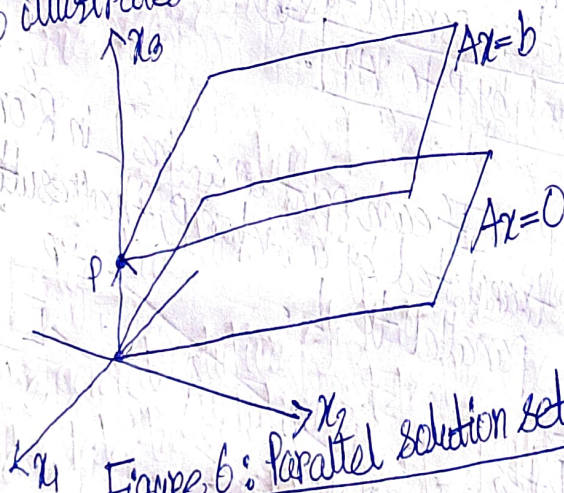


Figure 6: Parallel solution sets of $Ax=b$ and $Ax=0$

Writing a Solution Set (of a Consistent System) in Parametric Vector Form

1. Row reduce the augmented matrix to reduced echelon form.
2. Express each basic variable in terms of any free variables appearing in an equation.
3. Write a typical solution x as a vector whose entries depend on the free variables, if any.